When we do long division of $f \in k[x_1,\dots,x_n]$ by $f_1,\dots,f_s \in k[x_1,\dots,x_n]$, we maintain the following equality:

$$\underbrace{a_1}_{\text{emerging quotients}} f_1 + \dots + \underbrace{a_s}_{\text{emerging quotients}} f_s + \underbrace{p}_{\text{still to be divided by } f_1,\dots,f_s} + \underbrace{r}_{\text{emerging remainder}} = f \qquad (*)$$

1

Discussion area

xy^2 $\quad$ x^4y^3

x^2y $\quad$ x^1y^2

$$\frac{x^4y^3}{x^2y} = x^2y^2$$

Quotients_updated()

```
for i = 1,...,s:
    if LT(p) is divisible by LT(f_i):
        T := LT(p) / LT(f_i)
        a_i := a_i + T
        p := p - T f_i
        return True
return False
```

Multivariate_Division$(f, f_1,\dots,f_s)$:

Assumption: $f, f_1,\dots,f_s \in k[x_1,\dots,x_n]$, $f_i \neq 0$ for $i=1,\dots,s$.

Result: quotients and the remainder of the multivariate division of f by $f_1,\dots,f_s$.

```
r := 0
p := f                ▷ all terms of f to be divided
a_i := 0 for all i=1,...,s
▷ (*) is our invariant.
While p ≠ 0:
    if not Quotients_updated():
        r := r + LT(p)    ▷ LT depends on your order.
        p := p - LT(p)
Return a_1,...,a_s, r
```

2.2.1. Example. Lex-order. Divide $f = \underline{x^2y} + xy^2 + y^2$

by $f_1 = \underline{xy} - 1$ and $f_2 = \underline{y^2} - 1$.

2

(1)	$p = \underline{x^2y} + xy^2 + y^2$	$r = 0$	$a_1 = 0$	$a_2 = 0$
(2)	$p = \underline{xy^2} + x + y^2$	$r = 0$	$a_1 = x$	$a_2 = 0$
(3)	$p = \underline{x} + y^2 + y$	$r = 0$	$a_1 = x + y$	$a_2 = 0$
(4)	$p = \underline{y^2} + y$	$r = x$	$a_1 = x + y$	$a_2 = 0$
(5)	$p = \underline{y} + 1$	$r = x$	$a_1 = x + y$	$a_2 = 1$
(6,7)	$p = 0$	$r = x + y + 1$	$a_1 = x + y$	$a_2 = 1.$

2.2.2. Theorem Let $>$ be a monomial order on $\mathbb{Z}^n_{\geq 0}$ and let $f_1 \dots f_s \in k[x_1,\dots,x_n] \setminus \{0\}$.
Then every $f \in k[x_1, \dots, x_n]$ admits a representation $f = a_1 f_1 + \dots + a_s f_s + r$
where $a_1, \dots, a_s, r \in k[x_1,\dots,x_n]$ have the following properties:

3.

- No term of r is divisible by the leading term of f_i (for every $i = 1 \dots s$)
- $\mathrm{mdeg}(f) \geq \mathrm{mdeg}(a_i f_i)$ (for every $i = 1, \dots s$).

Proof: Let $a_1 \dots a_s, r$ be the result of the multivariate division algorithm. If the algorithm terminates, then its invariant guarantees that $f = a_1 f_1 + \dots + a_s f_s + r$. The multivariate division algorithm <u>does</u> terminate, because the leading term of p in this algorithm gets smaller w.r.t the order $>$ in each iteration. So, the respective monomials form a descending chain, which we know to be finite.
By the construction of the algorithm only the terms in p that are not divisible by $LT(f_1), \dots, LT(f_s)$ go into the remainder.
By construction when we add term T to a_i we have $\mathrm{mdeg}(T \cdot a_i) = \mathrm{mdeg}(p)$ for the current p, but we start with $p = f$ and $\mathrm{mdeg}(p)$ gets smaller in each iteration $\Rightarrow \mathrm{mdeg}(f) \geq \mathrm{mdeg}(a_i f_i)$. $\square$

2.2.3 Example. We want to test if $f \in \langle f_1, .., f_s \rangle$ using multivariate division, but it turns out that not every system of generators $f_1, .., f_s$ is good for this task. Let's have a bad example. Lex-order.

$f = \underline{xy^2} - x \qquad f_1 = x\underline{y} + 1, \qquad f_2 = \underline{y^2} - 1$

(1) $p = \underline{xy^2} - x \qquad r = 0 \qquad a_1 = 0 \qquad a_2 = 0$

(2) $p = \underline{-x} - y \qquad r = 0 \qquad a_1 = y \qquad a_2 = 0$

(3) $p = -\underline{y} \qquad r = -x \qquad a_1 = y \qquad a_2 = 0$

(4) $p = 0 \qquad r = -x - y \qquad a_1 = y \qquad a_2 = 0$

The result of the multivariate division algo.

non-zero remainder!

HOWEVER

$0 \cdot (xy+1) + x \cdot (y^2 - 1) = xy^2 - x$

2.3. Monomial ideals and Dickson's Lemma.

2.3.1 Definition Let $\langle f_s : s \in S \rangle$ denote the ideal generated by $f_s \in k[x_1, \ldots, x_n]$ with $s \in S$, where S is an index set (possibly infinite). This ideal consists of polynomials of the form

$$\sum_{s \in S} h_s f_s$$ where $h_s \in k[x_1, \ldots, x_n]$ and $h_s \neq 0$ for only finitely many choices of s

(Clearly, this is an ideal).

2.3.2 Def. Let $A \subseteq \mathbb{Z}^n_{\geq 0}$ (possibly infinite). Then, we call $\langle x^\alpha : \alpha \in A \rangle$ a monomial ideal.

2.3.3. Def. We introduce the partial order $\geq$ on $\mathbb{Z}^n_{\geq 0}$ by componentwise comparison

That is, $\alpha = (\alpha_1, \ldots, \alpha_n)$ and $\beta = (\beta_1, \ldots, \beta_n)$ satisfy $\alpha \geq \beta$ iff $\alpha_i \geq \beta_i$ for every $i = 1, \ldots, n$.

For example. $(4,3) \geq (2,1)$

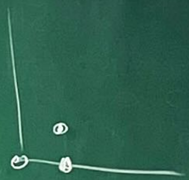
$(2,1) \not\geq (1,2)$

$(1,2) \not\geq (2,1)$

partial $\geq$ total monomial order $>$

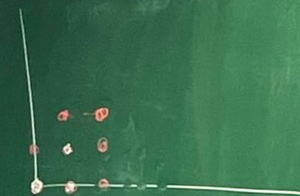
2.3.4. Definition. The Minkowski sum of A and B, where $A, B \subseteq \mathbb{Z}^n_{\geq 0}$, is $A + B = \{\alpha + \beta : \alpha \in A, \beta \in B\}$

6



$A = \{(0,0), (1,0), (1,1)\}$

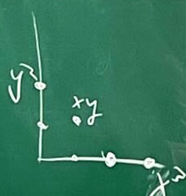
$B = \{(0,0), (1,0), (0,1), (1,1)\}$



$A + B$

2.3.5 Lemma. For $A \subseteq \mathbb{Z}^n_{\geq 0}$, one has

$$\langle x^\alpha : \alpha \in A \rangle = \mathrm{lin}_k \{ x^\beta : \beta \in A + \mathbb{Z}^n_{\geq 0} \}$$



$\langle x^3, xy, y^2 \rangle$